

Production and Costs

Chapter 21

All About Costs

- An **explicit cost** is a cost that is incurred when an actual payment is made.
- An **implicit cost** represents the value of resources used in production for which no actual payment is made. This is incurred as a result of a firm using its own resources that it owns or that the owners of the firm contribute to it.

Types of Profit

- **Accounting Profit** is the difference between total revenue and explicit costs.
- **Economic Profit** is the difference between total revenue and total opportunity cost, including both its explicit and implicit components.



Zero Economic Profit?

- It is possible for a firm to earn both a positive accounting profit and a zero economic profit.
- A firm that makes zero economic profit is said to be earning normal profit.
- Zero economic profit, the level of profit necessary to keep resources employed in a firm. A firm that earns normal profit is earning revenue equal to its total costs (explicit plus implicit costs).

Sunk Cost

Sunk cost is a cost incurred in the past that cannot be changed by current decisions and therefore cannot be recovered.

Fixed & Variable Inputs & Costs

- A fixed input is an input whose quantity cannot be changed as output changes in the short run.
- The costs associated with fixed inputs are fixed costs. A fixed cost doesn't change as output changes.
- A variable input is an input whose quantity can be changed as output changes in the short run.
- The costs associated with variable inputs are variable costs. A variable cost changes as output changes.

Short Run & Long Run

- The short run is a period in which some inputs are fixed.
- The long run is a period in which all inputs can be varied.
- The **Total Cost** is the sum of fixed and variable costs

Periods of Production, Inputs, and Costs

Short Run

INPUTS USED

1. Fixed
2. Variable

COSTS ASSOCIATED WITH INPUTS

1. Fixed Costs
2. Variable Costs

DEFINITION OF COSTS

1. Fixed Costs: Costs that do not change as output changes.
2. Variable Costs: Costs that change as output changes.

Long Run

INPUTS USED

1. Variable

COSTS ASSOCIATED WITH INPUTS

1. Variable Costs

DEFINITION OF COSTS

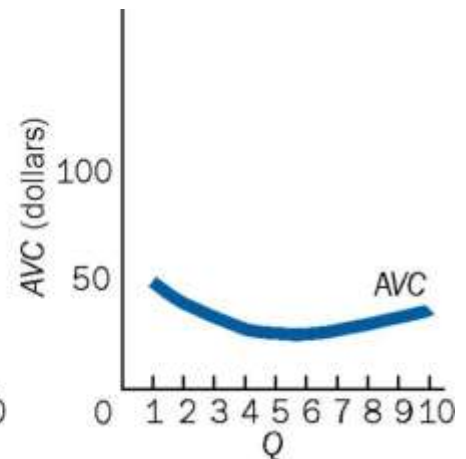
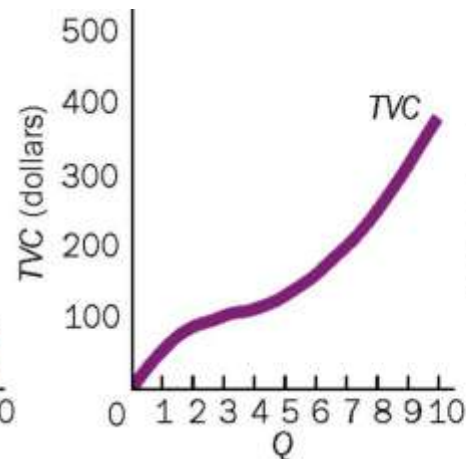
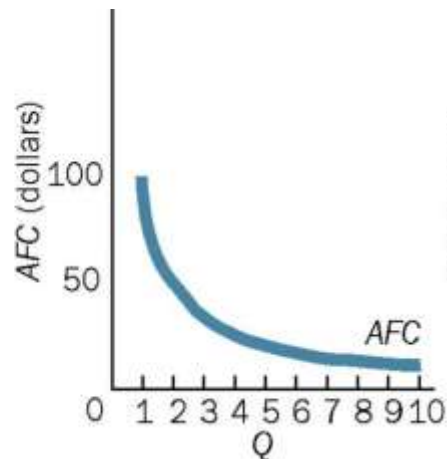
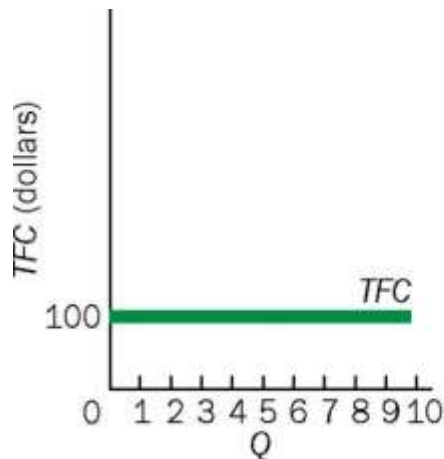
1. Variable Costs: Costs that change as output changes.

Computing and Graphing Total, Average and Marginal Cost Curves

- The Average Fixed Cost (AFC) = Total fixed cost divided by quantity of output
- The Average Variable Cost (AVC) = Total variable cost divided by quantity of output.
- The Average Total Cost (ATC) or Unit Cost= Total Cost divided by quantity of output.
- The Marginal Cost is the change in total cost or total variable cost that results from a change in output

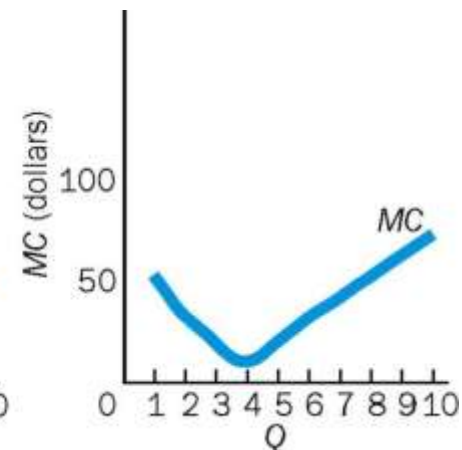
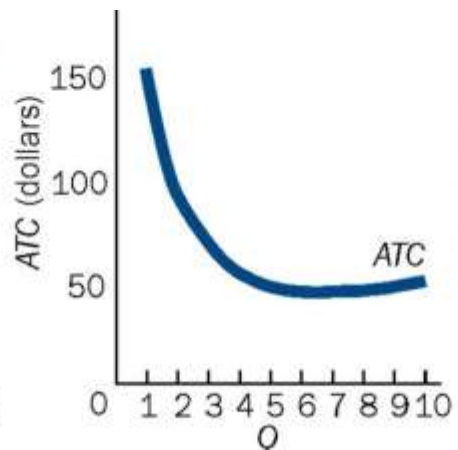
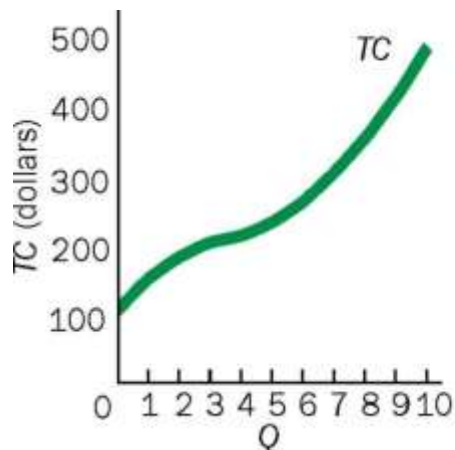
Total, Average, and Marginal Costs

(1) Quantity of Output, Q (units)	(2) Total Fixed Cost (TFC)	(3) Average Fixed Cost (AFC) $AFC = TFC/Q$ $= (2)/(1)$	(4) Total Variable Cost (TVC)	(5) Average Variable Cost (AVC) $AVC = TVC/Q$ $= (4)/(1)$
0	\$100	—	\$ 0	—
1	100	\$100.00	50	\$50.00
2	100	50.00	80	40.00
3	100	33.33	100	33.33
4	100	25.00	110	27.50
5	100	20.00	130	26.00
6	100	16.67	160	26.67
7	100	14.28	200	28.57
8	100	12.50	250	31.25
9	100	11.11	310	34.44
10	100	10.00	380	38.00



Total, Average, and Marginal Costs (Continued)

(6) Total Cost (TC) $TC = TFC + TVC$ $= (2) + (4)$	(7) Average Total Cost (ATC) $ATC = TC/Q$ $= (6)/(1)$	(8) Marginal Cost (MC) $MC = \Delta TC / \Delta Q$ $= \Delta(6) / \Delta(1)$, or $= \Delta TVC / \Delta Q$ $= \Delta(4) / \Delta(1)$
\$100.00	—	—
150.00	\$150.00	\$50.00
180.00	90.00	30.00
200.00	66.67	20.00
210.00	52.50	10.00
230.00	46.00	20.00
260.00	43.33	30.00
300.00	42.86	40.00
350.00	43.75	50.00
410.00	45.56	60.00
480.00	48.00	70.00



The Law of Diminishing Marginal Return

- As more and more variable inputs, such as labor and capital, were added to a fixed input, such as land, the variable inputs would yield smaller and smaller additions to output
- As ever larger amounts of a variable input are combined with fixed inputs, eventually the marginal physical product of the variable input will decline.
- The marginal physical product of a variable input is equal to the change in output that results from changing the variable input by one unit, holding all other inputs fixed.

The Law of Diminishing Returns

(1) Variable Input, Labor (workers)	(2) Fixed Input, Capital (units)	(3) Quantity of Output, Q (units)	(4) Marginal Physical Product of Variable Input (units) $\Delta(3)/\Delta(1)$
0	1	0	18
1	1	18	19
2	1	37	20
3	1	57	19
4	1	76	18
5	1	94	17
6	1	111	16
7	1	127	10
8	1	137	-4
9	1	133	-8
10	1	125	

More on Diminishing Marginal Returns

- If the law of diminishing marginal returns did not hold, then it would be possible to continue to add additional units of a variable input to a fixed input, and the marginal physical product of the variable input would never decline. We could increase output indefinitely as long as we continued to add units of a variable input to a fixed input.
- The law of diminishing marginal returns says that as more units of the variable input are added, each one has fewer units of the fixed input to work with; consequently, output rises at a decreasing rate.

The Link between MPP and Marginal Cost

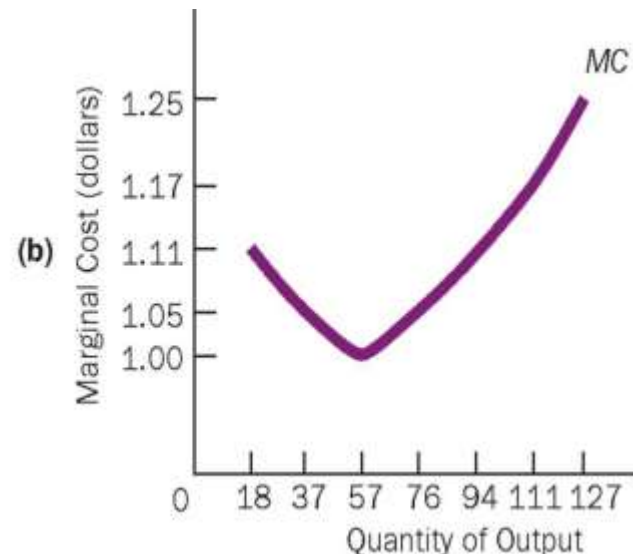
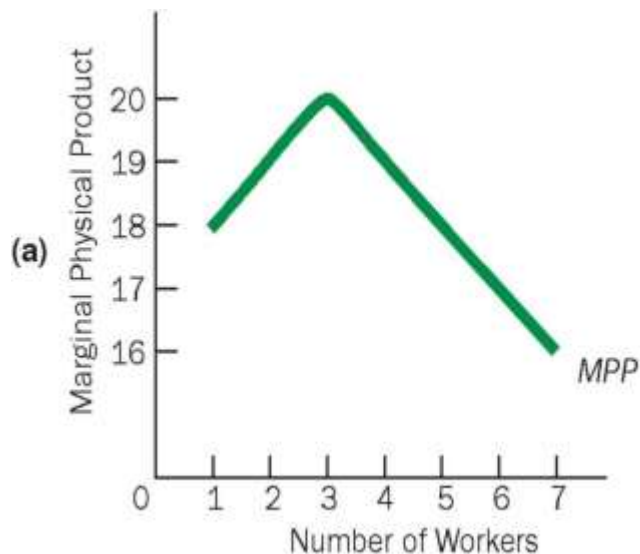
- Marginal cost is a reflection of the marginal physical product of the variable input.
- As the marginal physical product curve rises, the marginal cost curve falls; and as the marginal physical product curve falls, the marginal cost curve rises.
- What the marginal cost curve looks like depends on what the marginal physical product curve looks like.
- $MC = \Delta TC / \Delta Q$; $MC = W / MPP$; so,

$$\frac{W}{MPP \uparrow} = MC \downarrow$$

$$\frac{W}{MPP \downarrow} = MC \uparrow$$

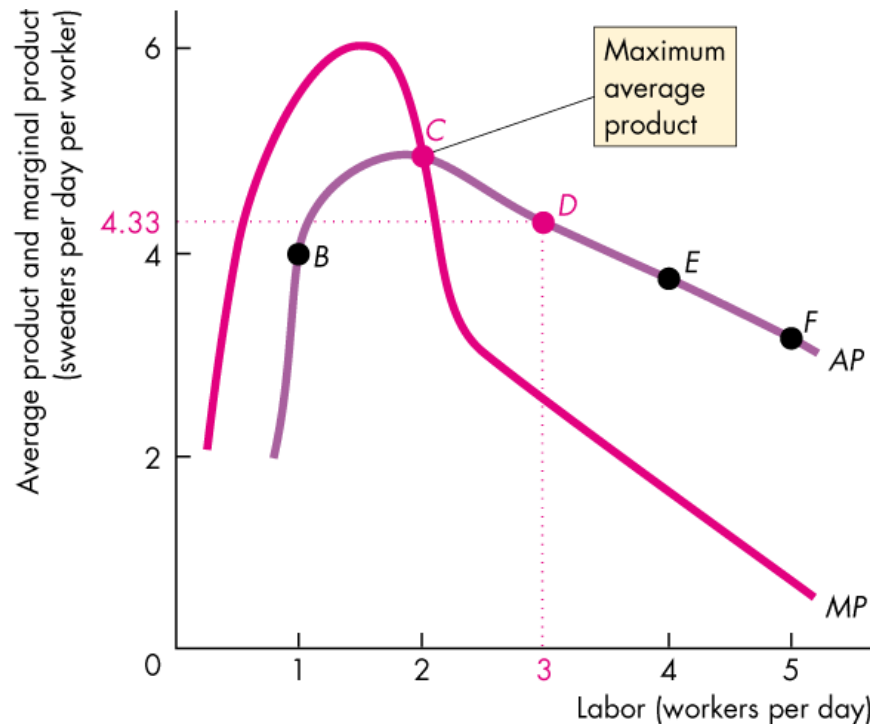
Marginal Physical Product and Marginal Cost

(1) Variable Input, Labor (workers)	(2) Fixed Input, Capital (units)	(3) Quantity of Output, Q (units)	(4) Marginal Physical Product of Variable Input (units) $\Delta(3)/\Delta(1)$	(5) Total Fixed Cost (dollars)	(6) Total Variable Cost (dollars)	(7) Total Cost (dollars) (5) + (6)	(8) Marginal Cost (dollars) $\Delta(7)/\Delta(3)$ or $\Delta(6)/\Delta(3)$
0	1	0		\$40	\$ 0	\$ 40	
1	1	18	18	40	20	60	\$1.11
2	1	37	19	40	40	80	\$1.05
3	1	57	20	40	60	100	\$1.00
4	1	76	19	40	80	120	\$1.05
5	1	94	18	40	100	140	\$1.11
6	1	111	17	40	120	160	\$1.17
7	1	127	16	40	140	180	\$1.25



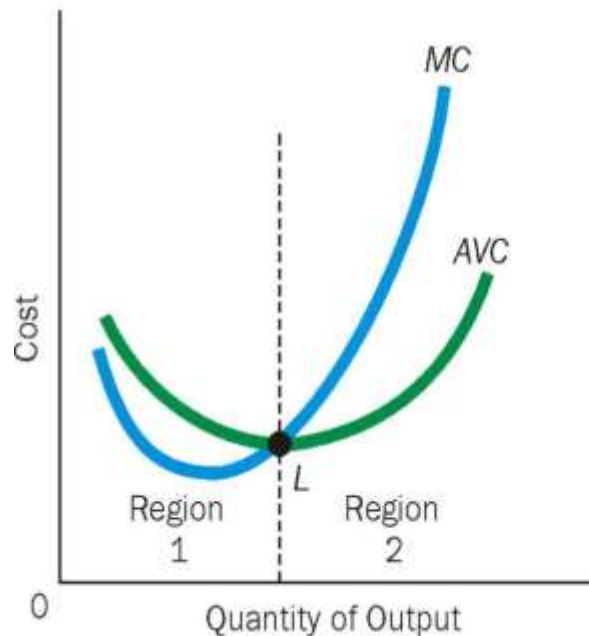
Average Productivity and Marginal Productivity

- Average physical product is output divided by the quantity of labor.
- When marginal product *is below* average product, average product decreases.
- When marginal product equals average product, average product is at its maximum.

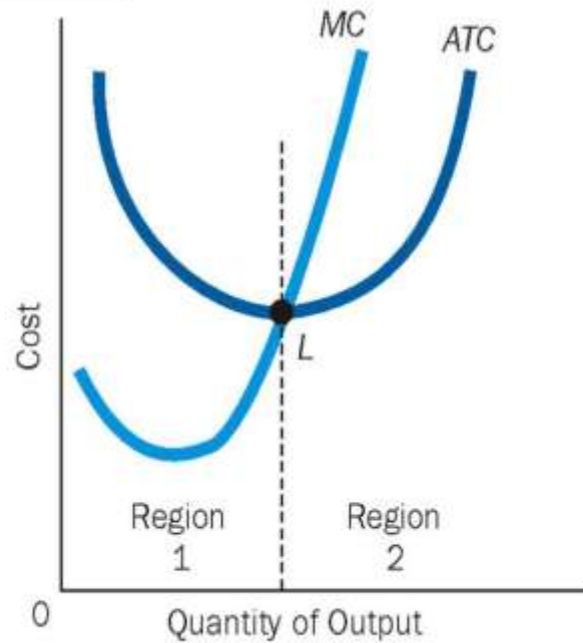


Average and Marginal Cost Curves

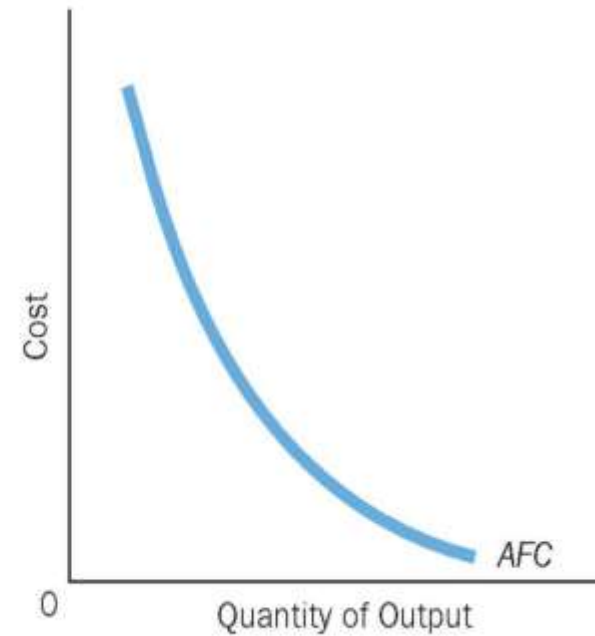
MC curve cuts both AVC and ATC curves at their respective low points.



(a)

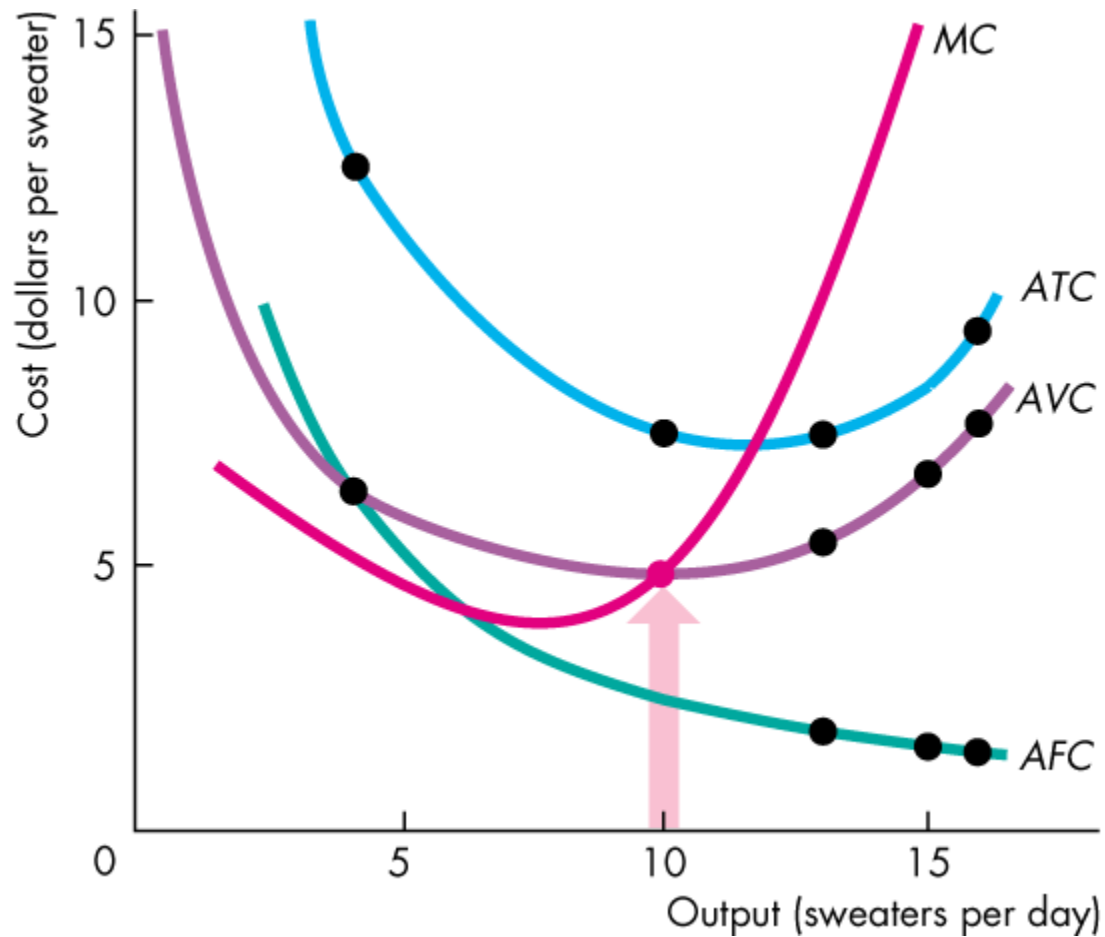


(b)



(c)

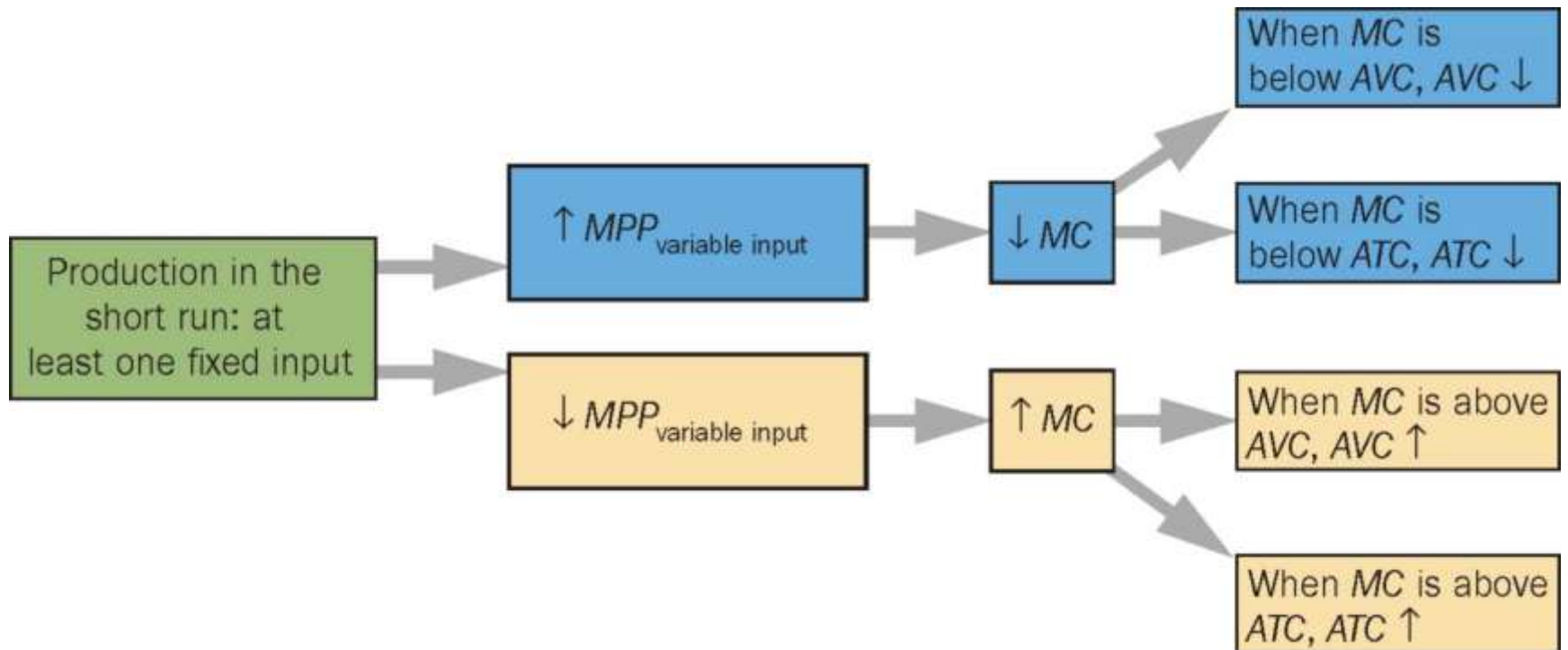
Short run cost curves



Tying Production To Costs

- Production underlies what many of the various cost curves look like.
- What happens in terms of production affects Marginal Cost, which in turn eventually affects Average Variable Cost and Average Total Cost.

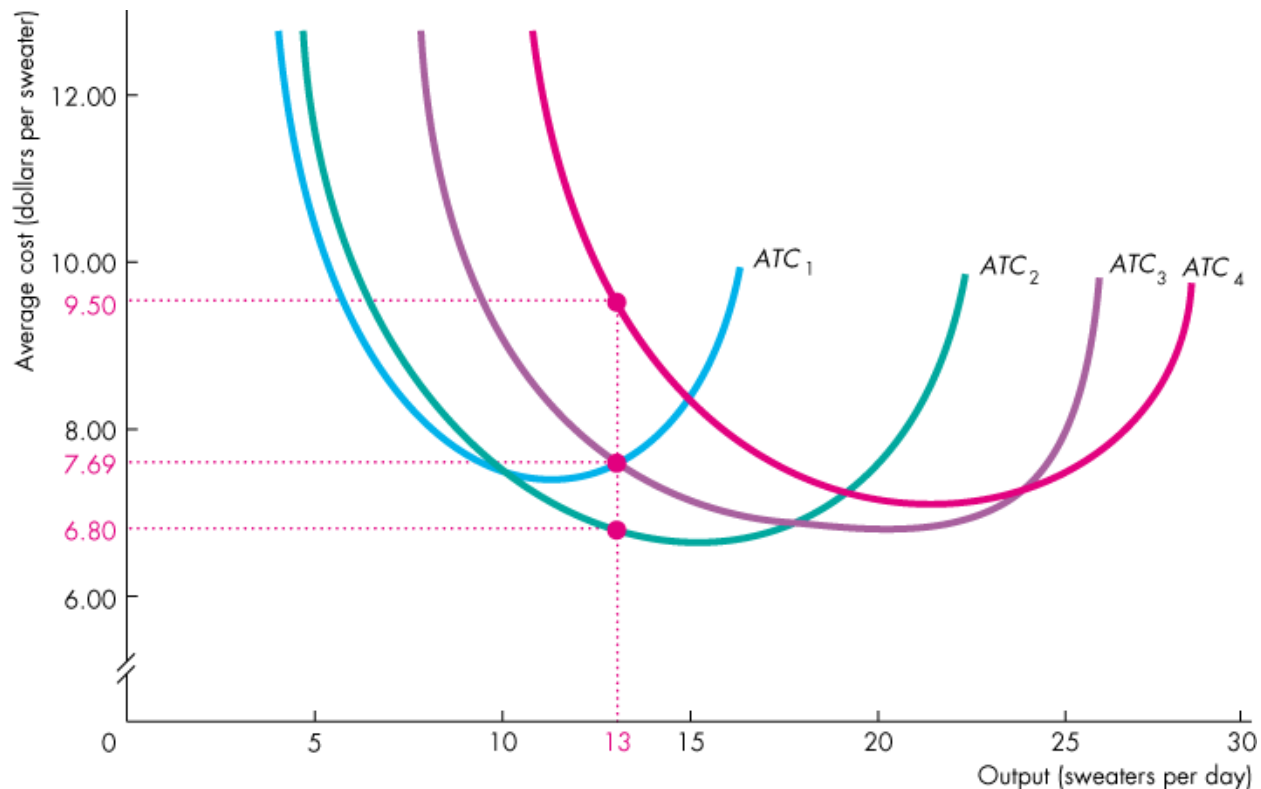
Tying Products to Costs



Short run and long run cost

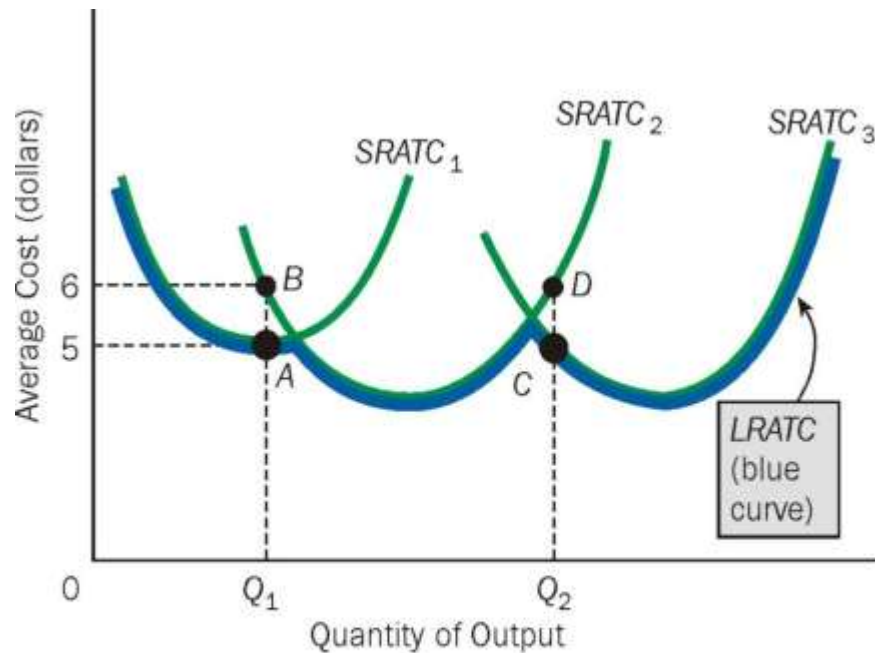
The average cost of producing a given output varies and depends on the firm's plant size.

The larger the plant size, the greater is the output at which ATC is at a minimum.

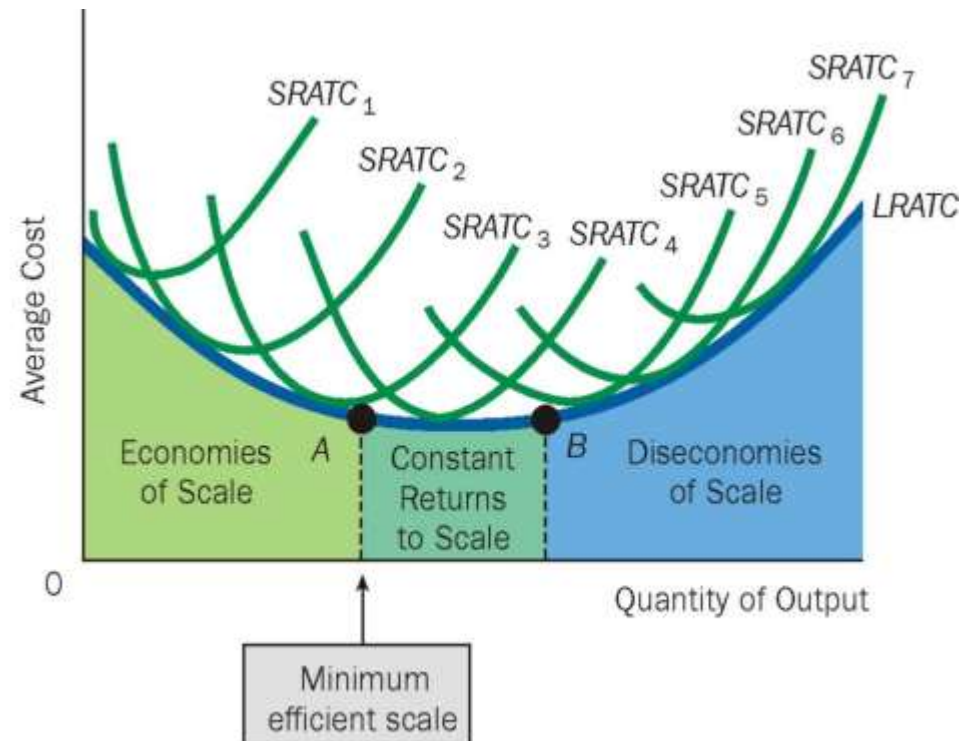


Long – Run Average Total Cost Curve

- Long Run Average Total Cost Curve shows the lowest unit cost at which the firm can produce any given level of output.



(a)



(b)

Economies of Scale, Diseconomies of Scale, and Constant Returns to Scale

- Economies of Scale exist when inputs are increased by some percentage and output increases by a greater percentage, causing unit costs to fall.
- Constant Returns to Scale exist when inputs are increased by some percentage and output increases by an equal percentage, causing unit costs to remain constant.
- Diseconomies of Scale exist when inputs are increased by some percentage and output increases by a smaller percentage, causing unit costs to rise.
- Minimum Efficient Scale is the lowest output level at which average total costs are minimized.

Up to a Certain Point, Long – Run unit Costs of Production Fall as a Firm Grows.

- **Economies of scale arise** as growing firms can offer greater opportunities for employees to specialize.
- Growing firms can take advantage of highly efficient mass production techniques and equipment that ordinarily require large setup costs and thus are economical only if they can be spread over a large number of units.
- **Diseconomies of scale usually arise** at the point where a firm's size causes coordination, communication, and monitoring problems.

Shifts In Cost Curves

- **Taxes:** affects variable costs. Because variable costs rise, total costs rise, and their average curves are shifted upward. Because marginal cost is the change in total cost divided by the change in output, marginal cost rises.
- **Input Prices:** A rise or a fall in variable input prices causes a corresponding change in the firm's average total, variable, and marginal cost curves.
- **Technological changes** often bring either the capability of using fewer inputs to produce a good or lower input prices.